6th Grade Accelerated Unit 13 Assessment Guide

General Information	The benchmarks on this assessment have assessment limits that may be helpful for the teacher to understand. The assessment limits for MA.6.AR.1.4 are "Items must include an expression with only one or two variables where coefficients, factors, and terms are integers only. Items will require the student to use one, all, or a combination of the associative property, distributive property, commutative property, or arithmetic operations to generate equivalent expressions." Teachers should note that the benchmark nor the assessment limits state that students should know when a property has been applied but that the items may require that the student employ one or more of the properties. Therefore, questions should not ask for a property name. The difficulty of the questions for MA.6.AR.1.4 are limited by the first assessment limit which limits the expression to one or two variables and all coefficients, factors, and terms are integers only. In grade 5, students evaluated multi-step numerical expressions using order of operations, MA.5.AR.2.2, where expressions were limited to no more than three operations that involve combinations of the four arithmetic operations and parentheses with whole numbers, decimals and fractions but could not include exponents or nested grouping symbols. Teachers should note that grade 6 students would be expected to answer questions for MA.6.AR.1.3 that go beyond what a 5th grader is expected to be able to do. This can be achieved by having questions that include integers, exponents, and/or nested grouping symbols. The questions on this assessment for MA.6.AR.1.4 includes at least one of these. For questions that assess MA.6.AR.1.1, MA.7.AR.1.1, and MA.7.AR.1.2, students may mistakenly write the variable as x instead of the given variable. Consider a small deduction of points if a student consistently makes this error throughout the assessment, but this should not be overemphasized at this level. Consider providing feedback to students if making this error rather than deducting points. The focus should be on the student's ability to solve the equations.
Calculator Information	**All questions are no calculator.** Fluency of operations with rational numbers is an important outcome of Grade 6 Accelerated. If a calculator is not provided for this assessment, then teachers should consider how to monitor simple calculation errors. If teachers want to consider student fluency of rational numbers, then if a student makes a simple calculation error that is carried through their work that the students receive partial credit.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.6.AR.1.3
☐ -22 ☐ -18 ☐ $(-2) - 20$ ☐ $(-2) + -20$ ☐ $2(-1) + 4(5)$ ☐ $2(-1) - 4(5)$		**Recommended Scoring** **(12 Total Points)**	Consider giving students 3 points for each correct choice. Consider deduction of all points for students who select all since the answers of -22 and -18 both could not be true.
Question	**2**	**Benchmark(s)**	MA.6.AR.1.1
Expression | $300 - 4h$		**Recommended Scoring** **(5 Total Points)**	Students may use a different variable, but it's recommended not to deduct points for this error. Consider giving students 2 points for $300 - h$.
Question	**3**	**Benchmark(s)**	MA.6.AR.1.1
☐ fourteen less than y ☐ y less than fourteen ☐ the sum of y and -14 ☐ y subtracted from -14 ☐ the difference between y and 14		**Recommended Scoring** **(9 Total Points)**	Consider giving students 3 points for each correct choice. Consider deduction of 3 points for each incorrect selection.
Question	**4**	**Benchmark(s)**	MA.6.AR.1.4
☐ $2x + 18y$ ☐ $18y - 6x$ ☐ $18y - x + 3x$ ☐ $9(2y - x) + 3x$ ☐ $-2(3x - 8y) - 2y$		**Recommended Scoring** **(6 Total Points)**	Consider giving students 3 points for each correct choice. Consider deduction of 2 points for each incorrect selection.

Question	5	Benchmark(s)	MA.6.AR.1.3
-4		Recommended Scoring (6 Total Points)	Consider giving students 3 points for correct substitution. Watch for the common error of writing 10^3 as 30.
Question	**6**	**Benchmark(s)**	MA.6.AR.1.4
☐ $2h + 33k$ ☐ $14h + 33k$ ☐ $8(3k + h) + 3(3k - 2h)$ ☐ $(8h - 6h) + (9k + 24k)$ ☐ $3(8k + 3k) - 2(-2h)$		Recommended Scoring (9 Total Points)	Consider giving students 3 points for each correct choice. Consider deduction of 2 points for each incorrect selection. Consider deduction of all points for students who select all since $2h + 33k$ and $14h + 33k$ as both could not be true.
Question	**7**	**Benchmark(s)**	MA.6.AR.1.1
Expression | $\frac{n}{8}$		Recommended Scoring (5 Total Points)	Students may write the answer as $n \div 8$.

Question	8	Benchmark(s)	MA.6.AR.2.1
A. -7		**Recommended Scoring** **(3 Total Points)**	Consider not giving any partial credit.
Question	**9**	**Benchmark(s)**	MA.6.AR.2.1
☐ -4 ☐ -2 ☐ -1 ☐ 0 ☐ 2 ☐ 5		**Recommended Scoring** **(9 Total Points)**	Consider giving students 3 points for each correct choice. Consider deduction of 3 points for each incorrect selection.
Question	**10a**	**Benchmark(s)**	MA.7.AR.1.1
$3x + 12$		**Recommended Scoring** **(6 Total Points)**	Students may incorrectly add x and $2x$. Each term can be considered as separate responses. Suggested points for each term is **3** points.
Question	**10b**	**Benchmark(s)**	MA.7.AR.1.1
$4k + 7$		**Recommended Scoring** **(6 Total Points)**	Each term can be considered as separate responses. Suggested points for each term is **3** points.
		Additional Notes	Consider student’s work for partial credit. Consider student’s work for partial credit. Consider deducting 1 point for students who make a minor calculation error.

Question	10c	Benchmark(s)	MA.7.AR.1.1
$2.9r + 8.5$		Recommended Scoring (6 Total Points)	Each term can be considered as separate responses. Suggested points for each term is **3** points.
		Additional Notes	Consider student's work for partial credit. Consider deducting 1 point for students who make a minor calculation error. A common misconception is not distributing the subtraction sign (-1) to both 6.3 and $-2.9r$.
Question	**11**	**Benchmark(s)**	MA.7.AR.1.1
$\frac{21}{8}w - \frac{28}{6}$		Recommended Scoring (6 Total Points)	Each term can be considered as separate responses. Suggested points for each term is **3** points.
		Additional Notes	Equivalent responses that use fractions should be accepted. From FLDOE's Grade 7 Big M, page 25, "For example, if students are given fractions, the solution should be demonstrated in fractions and not be converted to decimals." Consider deducting 1 point for students who make a minor calculation error.

Question	12a	Benchmark(s)	MA.7.AR.1.2
Answers will vary. Use of $m = 0$ will show equivalence and is the only value that should not be used.		Recommended Scoring (6 Total Points)	Suggested points for choosing a value other than 0 and showing that the expressions are not equivalent is **6** points.
		Additional Notes	Consider student's work for partial credit. Consider deducting 1 point for students who make a minor calculation error.
Question	**12b**	**Benchmark(s)**	MA.7.AR.1.2
$28m + 70 - 2$ $28m + 68$		Recommended Scoring (4 Total Points)	Suggested points is **4** points for work showing the expressions are not equivalent.
Question	**12c**	**Benchmark(s)**	MA.7.AR.1.2
The expressions ○ are / ○ are not equivalent.		Recommended Scoring (2 Total Points)	Suggested points is **2** points for correct choice in the statement.